Multifeature Domain Fusion Extraction of Buildings From Airborne L-Band CSAR Images Using Subaperture Image Scattering Characteristics

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***Abstract*—As one of the critical targets in urban areas, the detection and extraction of buildings in synthetic aperture radar (SAR) images are of paramount importance. Distinguishing artificial structures, such as buildings, from natural backgrounds through feature extraction serves as the foundation for subsequent interpretation. Unlike conventional linear SAR (LSAR), circular SAR (CSAR) enables multiaspect observation and can acquire target scattering information from various aspects, which facilitates the extraction of ground targets from the CSAR image. Low-frequency SAR systems (e.g., L-band and P-band) present greater challenges for building extraction due to their longer wavelengths and lower resolution. Consequently, low-frequency SAR is predominantly employed for concealed target imaging and detection, with limited applications in man-made feature extraction. To address these limitations, this article proposes a novel multifeature domain fusion-based method for building extraction in low-frequency CSAR images. Specifically, the method exploits two distinct characteristics of buildings in CSAR subaperture sequences: strong scattering and anisotropic scattering. For strong scattering, a spatial–frequency matrix decomposition-based strong scattering points extraction (SFMD-SSPE) method is proposed, which integrates frequency- and spatial-domain information to identify high-intensity scattering pixels across subapertures. For anisotropic scattering, a low-rank anisotropic scattering points extraction (LASPE) method is designed, leveraging the statistical properties of the aspect domain to isolate anisotropic scattering points. Experimental validation on airborne L-band measured data demonstrates that the proposed method achieves superior performance compared to existing CSAR feature extraction approaches. The fusion of multidomain features effectively enhances the accuracy and robustness of building extraction.**

***Index Terms*—Anisotropic scattering, building extraction, circular synthetic aperture radar (CSAR), multifeature fusion, strong scattering.**

I. Introduction

SYNTHETIC aperture radar (SAR) is extensively utilized across diverse fields owing to its all-day, all-weather operability and strong penetration capabilities [1]. As one of the most prevalent targets in SAR imagery, buildings play a critical role in digital city construction and disaster emergency response, making the accurate acquisition of their structural information essential [2], [3], [4], [5]. Current research on building extraction primarily relies on single-aspect SAR images [6], [7], with most spatial-feature-based detection methods leveraging composite characteristics such as layover, shadow, and double-bounce scattering in SAR data [8], [9] to focus on geometric structure extraction [10]. For instance, Simonetto et al. [11] identified buildings by matching bright lines in two images, though inherent speckle noise often impedes direct line detection. Zhao et al. [12] adopted a marker-controlled watershed transform for contour extraction, where internal markers were derived from constant false alarm rate (CFAR) detection, while external markers relied on energy ring detection: a method limited by the difficulty of accurately extracting shadow features. These approaches are highly susceptible to speckle noise and are typically confined to simple scenarios with clearly visible building edges and minimal shadow interference. Alternatively, frequency-domain methods like the wavelet transform and the Fourier transform require predefined basis functions (e.g., harmonic or wavelet bases), whereas bidimensional empirical mode decomposition (BEMD) [13] is entirely data-driven, adaptively decomposing data via iterative extraction of intrinsic mode functions (IMFs) to capture local oscillations. This eliminates subjective basis selection, better preserves edges and fine details, and proves particularly suitable for nonlinear and nonstationary data analysis.

The aforementioned method is widely employed for the interpretation of linear SAR (LSAR) images. However, the extraction performance is constrained by the limited observation angles of LSAR. Moreover, existing algorithms often consider features in a single domain independently, resulting in an incomplete characterization of the scattering properties of ground targets. With the continuous advancement of circular SAR (CSAR) imaging technology in recent years [14], [15], [16], [17], [18], algorithms for CSAR imagery have garnered significant research attention [19], [20], [21], [22], [23], [24], [25], [26]. Early approaches typically required fully polarimetric CSAR data. For instance, Li et al. [27] proposed an anisotropic target detection method based on likelihood ratio sorting and CFAR detection. Xue et al. [28] introduced the multiaperture polarization entropy (MAPE), which leverages anisotropic scattering characteristics of man-made targets for classification. Unlike conventional polarization

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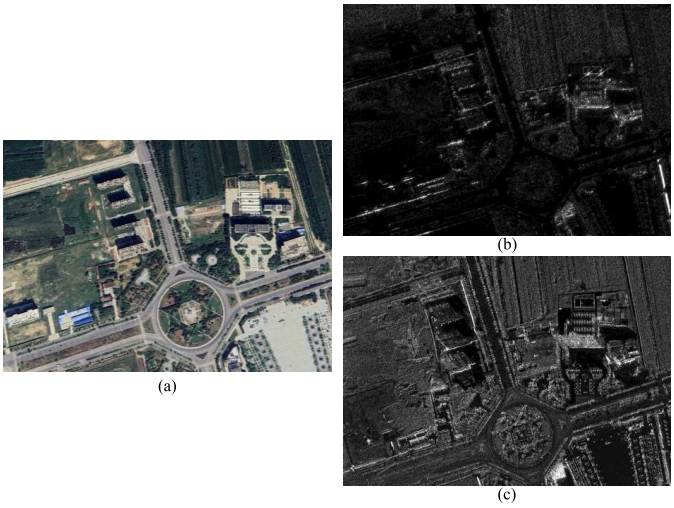


Fig. 1. Comparison of SAR images from different frequency bands. (a) Optical image. (b) Low-frequency SAR image (L-band). (c) High-frequency SAR image (Ku-band).

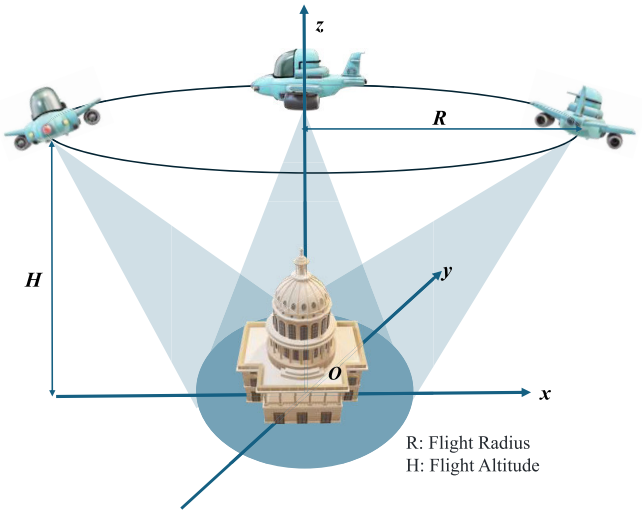


Fig. 2. CSAR imaging geometry.

TABLE I
RECENT ADVANCEMENTS IN CSAR-BASED FEATURE EXTRACTION TECHNIQUES

Methods	Proposed Time	CSAR data bands
The method in [20]	2019	W
The method in [31]	2023	X
The method in [33]	2021	C
The method in [35]	2019	C
The method in [36]	2020	C

entropy, MAPE integrates intensity variations across azimuth angles and polarization modes, enabling discrimination of both anisotropic targets and polarization-sensitive isotropic targets. Building on this, Tan et al. [29] achieved improved bridge extraction accuracy using MAPE. However, fully polarimetric CSAR data acquisition demands ideal experimental conditions, and the substantial computational load from four polarization channels poses practical challenges.

Subsequent studies demonstrated that single-polarization data could yield comparable results [30], [31], [32]. Inspired by thermodynamic entropy, Teng et al. [32] proposed aspect entropy to quantify target anisotropy using single-polarization echoes. Nevertheless, natural terrain retains anisotropy in low-frequency bands, complicating the distinction between natural and man-made targets in low-frequency CSAR imagery. Yue et al. [33] combined statistical distribution and membership degree features for C-band target extraction, though parameter estimation for statistical models remained problematic. Zhao et al. [34] developed a building contour extraction method integrating fuzzy C-means (FCMs) clustering with aspect entropy, achieving precise geometric measurements for Ku-band buildings. However, this method relies on manual thresholding for membership degrees, requiring empirical tuning of hyperparameters. Notably, Table I reveals that existing algorithms predominantly focus on single-feature domains and are primarily validated in high-frequency bands, which exhibit simpler noise distributions. This limitation is further illustrated in Fig. 1, where buildings in low-frequency SAR images show a higher tendency to be confused with natural features compared to their high-frequency counterparts—highlighting a significant research challenge in low-frequency building extraction.

This article analyzes the scattering model of buildings and proposes a novel building extraction algorithm for airborne L-band CSAR images by incorporating the scattering characteristics of man-made structures. The main contributions of this work are summarized as follows.

1) To address the strong scattering property of buildings, the spatial–frequency matrix decomposition-based strong scattering points extraction (SFMD-SSPE) method is proposed for CSAR subaperture images. The approach extracts salient feature points from frequency and spatial domains using BEMD and robust principal component analysis (RPCA), respectively, followed by solving the membership matrix with a fast filtered fuzzy C-means (F^3CM) to obtain normalized weights.
2) Considering the anisotropic scattering behavior of buildings as man-made structures, the low-rank anisotropic scattering points extraction (LASPE) method, based on the statistical characteristics of low-rank CSAR subaperture image sequences, is presented. A statistical scattering feature extraction module in the aspect domain is designed using mode–variance characteristics.
3) Buildings are extracted by jointly utilizing their strong scattering and anisotropic properties. Experimental validation on real L-band airborne CSAR data demonstrates the effectiveness of the proposed algorithm.

II. SCATTERING CHARACTERISTIC ANALYSIS OF BUILDINGS

As illustrated in Fig. 2, this schematic depicts the CSAR imaging geometry, where *xoy* represents the imaging ground

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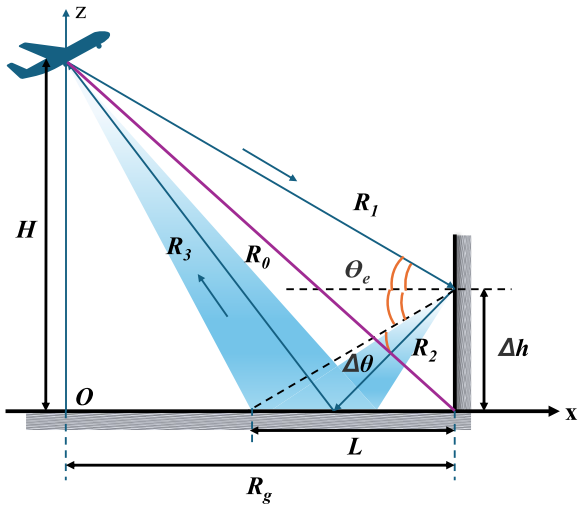


Fig. 3. Schematic of double-bounce scattering from buildings.

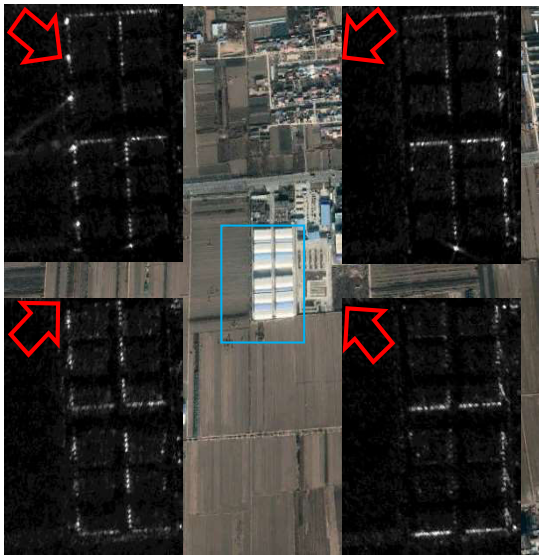


Fig. 4. Anisotropy of building subaperture image sequences. (The red arrow indicates the radar detection direction.)

plane and oz denotes the height direction. The airborne platform moves along a circular trajectory with radius R at a constant altitude H. Throughout the motion, the radar beam continuously points toward the central imaging area to acquire full-azimuth scattering information of the target [37].

A. Strong Scattering Characteristics

In CSAR subaperture images, the amplitude of double-bounce scattering is typically the most prominent feature in high-resolution SAR imagery. This scattering response generally corresponds to the edges of buildings, making it one of the most critical features for building extraction [34].

As illustrated in Fig. 3, the double-bounce scattering path can be temporally decomposed into three segments: R_1, R_2, and R_3 [16]. Assuming that the airborne radar antenna is positioned at altitude H with its nadir point at O, when the radar beam illuminates the building facade of height Δh through the path R_1. The first reflection exhibits an angular variation $\Delta\theta$ relative to specular reflection. Subsequently, the signal undergoes secondary reflection via the path R_2 on the ground before returning to the radar antenna through the path R_3. Notably, the lengths of R_1, R_2, and R_3 marked in the figure are not unique but vary with $\Delta\theta$. In addition, R_g is the ground distance from the radar to the building target. Based on the geometric relationships in the schematic, the following is derived:

$$\begin{cases} \theta_e = a\tan\left[\dfrac{H-\Delta h}{R_g}\right] \\ R_1 = \sqrt{(H-\Delta h)^2 + R_g^2} \\ R_2 = \dfrac{\Delta h}{\sin(\theta_e+\Delta\theta)} \\ L = \dfrac{\Delta h}{\tan(\theta_e+\Delta\theta)} \\ R_3 = \sqrt{(R_g-L)^2 + H^2} \\ R_0 = \sqrt{R_g^2+H^2}. \end{cases} \tag{1}$$

For airborne radar systems, the antenna altitude is typically significantly greater than the building height, yielding

$$R_1 + R_2 + R_3 \approx 2R_0. \tag{2}$$

This implies that the double-bounce scattering energy will be projected onto the building boundaries, manifesting as bright linear segments in single-angle SAR images. When performing full-aspect (0° ~ 360°) observation and imaging, these bright lines will form a closed polygon.

B. Subaperture-Dependent Anisotropic Scattering

In CSAR subaperture image sequences, the increased angular observation range causes building targets to lose isotropy. As shown in Fig. 4, the backscattering intensity of buildings varies at different azimuth angles. This anisotropy enables us to extract building features from multiangle CSAR image sequences.

III. Proposed Methodology

A. Framework of the Proposed Method

As mentioned earlier, in CSAR images, building targets not only exhibit strong scattering characteristics at certain angles (manifesting as observable bright lines due to double-bounce scattering) but also demonstrate anisotropy, where their scattering amplitude varies with observation angles following specific patterns.

A novel building extraction method based on CSAR imagery is proposed to leverage these properties. The method first applies filtering preprocessing to the experimentally acquired low-noise CSAR subaperture image sequence. It then separately extracts features related to both the strong scattering characteristics and the anisotropy of buildings. Finally, the extracted features are fused to obtain the building extraction results. Fig. 5 illustrates the workflow of the proposed algorithm.

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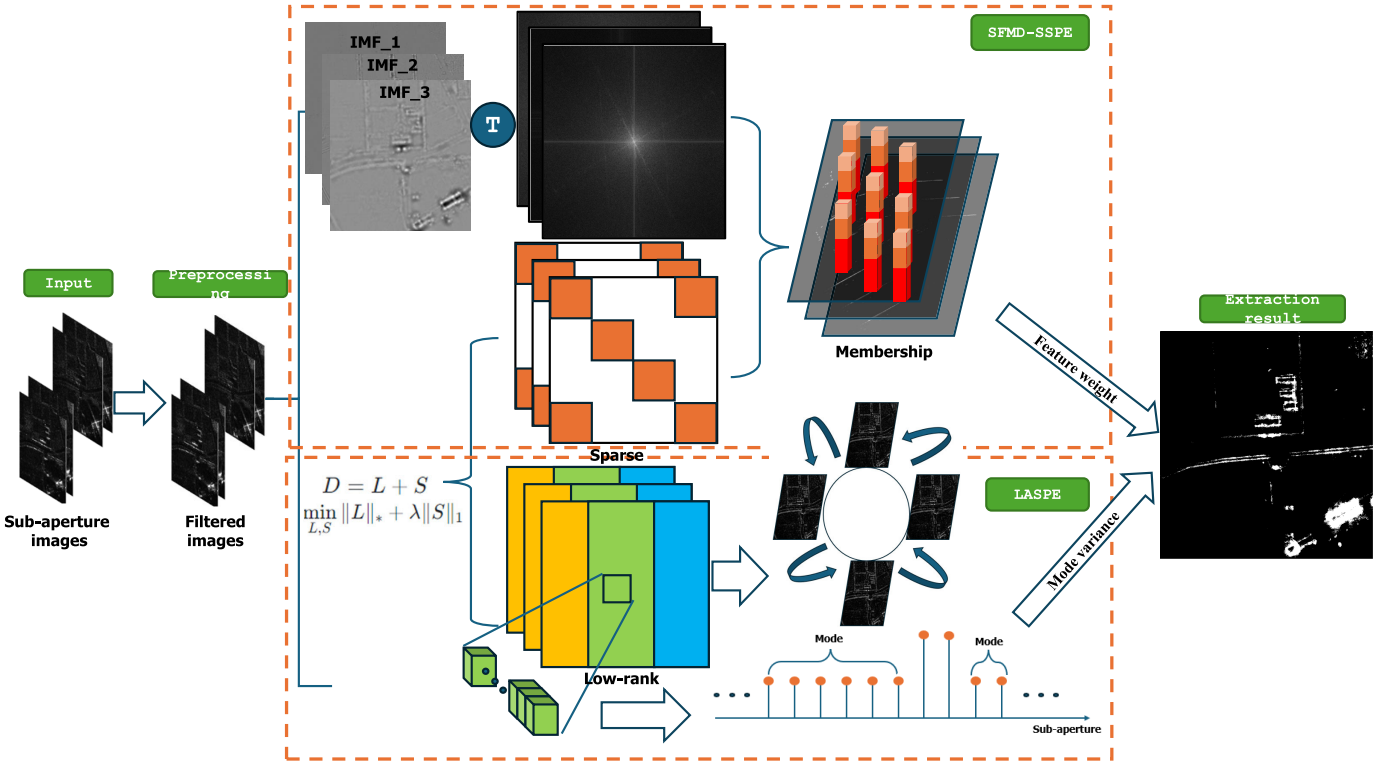


Fig. 5. Overall framework of the proposed algorithm.

Step 1: The first step involves dividing the full-aperture data into multiple subapertures. After imaging, each subaperture image undergoes prefiltering to suppress background noise, resulting in a filtered subaperture image sequence that serves as the input for the algorithm. The azimuth resolution of each subaperture image should be smaller than the azimuthal extent of the building to preserve its scattering characteristics [16]. The subaperture division criteria are as follows:

$$\rho_a = \frac{K_a \lambda}{4 \sin\left(\frac{\varphi}{2}\right)} < E \tag{3}$$

where ρ_a is the azimuth resolution, K_a is the weighting factor caused by subaperture weighting, λ is the wavelength, φ is the size of the subaperture angle, and E is the azimuthal extent of the building.

Step 2: The subaperture images undergo iterative BEMD to obtain the IMF matrix sequence. For each subaperture image, the first three IMF matrices are summed to extract the high-frequency components of the subaperture image sequence. Subsequently, RPCA is applied to decompose each subaperture image, yielding a sparse matrix. The two matrices are fused during F^3CM processing to derive the membership matrix, with normalized weights obtained as the result.

Step 3: The fused low-rank matrix sequence with normalized weights from the previous step is analyzed in the aspect domain to compute its dispersion. Specifically, the discrete degree is calculated by evaluating the statistical variance characteristics of each pixel in the subaperture sequence based on the mode, thereby determining the scattering intensity variation dispersion of each pixel across the subaperture sequence.

Step 4: The fused feature map obtained from the aforementioned steps is subjected to unsupervised classification using the multiple Otsu thresholding method to produce the final result.

B. BEMD-Based Feature Extraction in the Frequency Domain

Based on the analysis in Section II, buildings in CSAR subaperture images exhibit double-bounce scattering at their edges due to their geometric structures. Therefore, the proposed method employs BEMD to extract high-frequency components from the subaperture images. The key steps of the proposed method are as follows.

Step 1: The proposed BEMD-based decomposition process begins by initializing the residual $R_0(x, y, n)$ and intermediate component $d_0(x, y, n)$ with the original nth CSAR subaperture image $f(x, y, n)$.

Step 2: The ith IMF matrix is extracted through an iterative process involving extrema detection, envelope construction using interpolation, mean envelope calculation, residual component updating, and IMF condition verification.

First, local maxima and minima of the intermediate component $d_{k-1}(x, y, n)$ are detected through neighborhood comparison. Then, the upper envelope $u_k(x, y, n)$ and lower envelope $l_k(x, y, n)$ are constructed using interpolation during the kth iteration. Next, the mean envelope $m_k(x, y, n)$ is obtained by averaging the upper and lower envelopes. Subsequently, the residual component is updated by subtracting the mean envelope from the previous intermediate component. Finally, the resulting component $d_k(x, y, n)$ is evaluated against

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the IMF conditions to determine whether the decomposition process should terminate

$$m_k(x,y,n) = \frac{l_k(x,y,n) + u_k(x,y,n)}{2} \tag{4}$$

$$d_k(x,y,n) = f(x,y,n) - m_k(x,y,n) \tag{5}$$

$$\mathrm{SD} = \frac{\sum_{x,y}(d_k(x,y,n) - d_{k-1}(x,y,n))^2}{\sum_{x,y}(d_{k-1}(x,y,n))^2} 0. \tag{6}$$

If SD $< \epsilon$ (typically, ϵ is set between 0.2 and 0.3) or the maximum iteration count is reached, then $d_k(x,y,n)$ is recorded as IMF_i. If the condition is not satisfied, the process repeats for k iterations until IMF_i is obtained

$$d_k(x,y,n) = d_{k-1}(x,y,n) - m(x,y,n) \tag{7}$$

$$\mathrm{IMF}_i(x,y,n) = d_k(x,y,n). \tag{8}$$

Step 3: Update the residual matrix, then repeat Step 2 until convergence

$$R_i(x,y,n) = R_{i-1}(x,y,n) - \mathrm{IMF}_i(x,y,n). \tag{9}$$

Ultimately, the original CSAR subaperture image is decomposed into multiple IMF matrices and a residual matrix

$$f(x,y,n) = R(x,y,n) + \sum \mathrm{IMF}_i(x,y,n). \tag{10}$$

As illustrated in Fig. 6, for CSAR subaperture imagery, high-frequency components are predominantly concentrated within the first three IMF matrices. Consequently, our algorithm exclusively employs the summation of these initial three IMF matrices as the extraction result for feature weights of strong scattering points in the image

$$M_{\mathrm{imf}}(x,y,n) = \sum_{i=1}^{3} \mathrm{IMF}_i(x,y,n). \tag{11}$$

To elucidate the implementation details of the frequency-domain feature extraction method, Algorithm 1 presents the pseudocode for the proposed approach.

Algorithm 1 BEMD-Based Feature Extraction in the Frequency Domain

Require: $f(x,y,n)$: The n-th CSAR sub-aperture image ϵ: Standard deviation threshold (typically 0.2–0.3) max_iter: Maximum iteration count

Ensure: $M_{\mathrm{imr}}(x,y,n)$: Summation of first three IMF matrices

$R_0(x,y,n) \leftarrow f(x,y,n)$
$i \leftarrow 1$
while $i \le 3$ **do**
  $k \leftarrow 1$
  $d_0(x,y,n) \leftarrow R_{i-1}(x,y,n)$
  repeat
    $\mathcal{M}_{\max} \leftarrow \mathrm{findLocalMaxima}(d_{k-1}(x,y,n))$
    $\mathcal{M}_{\min} \leftarrow \mathrm{findLocalMinima}(d_{k-1}(x,y,n))$
    $u_k(x,y,n) \leftarrow \mathrm{surfaceInterpolation}(\mathcal{M}_{\max})$
    $l_k(x,y,n) \leftarrow \mathrm{surfaceInterpolation}(\mathcal{M}_{\min})$
    $m_k(x,y,n) \leftarrow \frac{u_k(x,y,n)+l_k(x,y,n)}{2}$
    $d_k(x,y,n) \leftarrow d_{k-1}(x,y,n) - m_k(x,y,n)$
    $SD \leftarrow \frac{\sum_{x,y}|d_{k-1}(x,y,n)-d_k(x,y,n)|^2}{\sum_{x,y}|d_{k-1}(x,y,n)|^2}$
    $k \leftarrow k+1$
  until $SD < \epsilon$ **or** $k >$ max_iter
  $\mathbf{IMF}_i(x,y,n) \leftarrow d_k(x,y,n)$
  $R_i(x,y,n) \leftarrow R_{i-1}(x,y,n) - \mathbf{IMF}_i(x,y,n)$
  $i \leftarrow i+1$
end while
$M_{\mathrm{imf}}(x,y,n) \leftarrow \sum_{j=1}^{3} \mathbf{IMF}_j(x,y,n)$
return $M_{\mathrm{imf}}(x,y,n)$

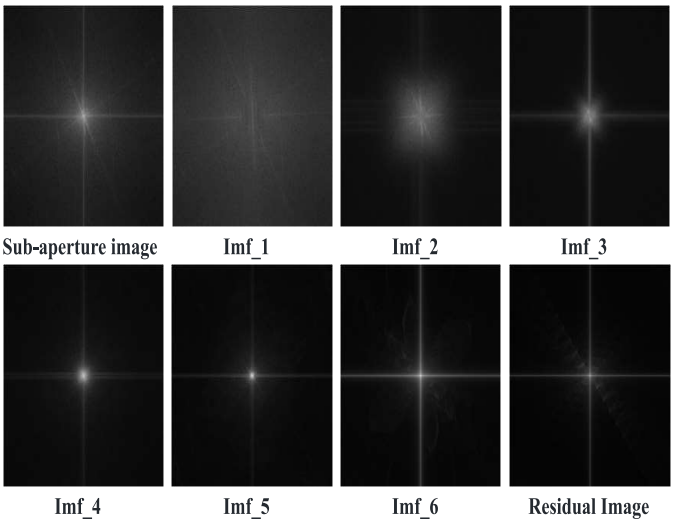


Fig. 6. Spectral diagrams of IMF matrix decomposition for a subaperture CSAR image.

C. RPCA-Based Feature Extraction in the Spatial Domain

In this study, the low-rank decomposition of multiaspect subaperture SAR images is employed to obtain the corresponding low-rank matrix and sparse matrix, thereby achieving spatial-domain feature extraction. The low-rank matrix can subsequently undergo further feature extraction in the aspect domain to identify anisotropic scattering pixels, while the sparse matrix typically contains strong scatterers from the subaperture images.

Let the decomposed low-rank matrix be denoted as $L(x,y,n)$ and the sparse matrix as $S(x,y,n)$. The decomposition is solved using RPCA [38].

Step 1: The RPCA problem decomposes a CSAR subaperture matrix into a low-rank matrix and a sparse matrix by solving

$$\min_{L,S} \|L(x,y,n)\|_* + \lambda \|S(x,y,n)\|_1 \tag{12}$$

$$f(x,y,n) = L(x,y,n) + S(x,y,n) \tag{13}$$

where $\|L(x,y,n)\|_*$ denotes the nuclear norm of matrix $L(x,y,n)$ (i.e., the sum of its singular values), which enforces the low-rank constraint. $\|S(x,y,n)\|_1$ represents the L_1-norm of matrix $S(x,y,n)$ (i.e., the sum of absolute values of its elements), imposing sparsity. The parameter $\lambda > 0$ is a balancing coefficient, typically set as $\lambda = (1/\sqrt{\max(r,c)})$, where r and c are the number of rows and columns of the CSAR subaperture image, respectively.

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Step 2: The augmented Lagrangian method (ALM) transforms this into

$$L_a(L,S,T,\mu) = \|L\|_* + \lambda\|S\|_1 + \langle T, f-L-S\rangle + \frac{\mu}{2}\|f-L-S\|_F^2 \tag{14}$$

where T is the Lagrange multiplier, $\mu > 0$ is a penalty parameter, $\langle\cdot,\cdot\rangle$ denotes the inner product, and $\|\cdot\|$ denotes the Frobenius norm.

In each iteration of the ALM, the matrix L is updated via singular value thresholding (SVT)

$$L_{k+1} = \arg\min_L \|L\|_* + \frac{\mu_k}{2}\|f-L-S_k+\mu_k^{-1}T_k\|_F^2. \tag{15}$$

The closed-form solution is given by

$$L_{k+1} = D_{\frac{1}{\mu_k}}\left(f - S_k + \frac{T_k}{\mu_k}\right) \tag{16}$$

where $D_{(1/\mu_k)}(\cdot)$ represents the SVT.

While the matrix L is updated via the soft thresholding operator

$$S_{k+1} = \arg\min_S \lambda\|S\|_1 + \frac{\mu_k}{2}\|f-L_{k+1}-S+\mu_k^{-1}T_k\|_F^2. \tag{17}$$

The closed-form solution is given by

$$S_{k+1} = E_{\frac{\lambda}{\mu_k}}\left(f - L_{k+1} + \frac{T_k}{\mu_k}\right) \tag{18}$$

where $E_{(\lambda/\mu_k)}(\cdot)$ represents the element-wise soft thresholding operator.

D. Mode–Variance-Based Feature Extraction in the Aspect Domain

Based on the preceding analysis, man-made structures exhibit anisotropic scattering characteristics under CSAR subapertures, which can be quantitatively characterized in the aspect domain. This study employs a mode-based variance statistic to quantify the dispersion of scattering amplitudes across different subapertures, thereby enabling further discrimination between natural terrain and man-made object pixels. This metric is conceptually analogous to conventional variance, defined mathematically as follows:

$$V(x,y) = \sum_{n=1}^{N}\left[L(x,y,n) - \text{Mode}(x,y)\right]^2 \tag{19}$$

where $\text{Mode}(x,y)$ denotes the mode of backscattered amplitude values across the subaperture sequence for the pixel at coordinates (x,y). N represents the length of the subaperture sequence. $L(x,y,n)$ indicates the amplitude value of the pixel at (x,y) in the nth low-rank subaperture image.

E. Weight Estimation of Strong Scattering Points in Joint Spatial–Frequency Domain

Following the extraction of strong scattering points in spatial and frequency domains (Sections III-A and III-B), their weights are computed via F^{3}CM clustering to obtain the subaperture membership matrix. As an unsupervised fuzzy clustering method, FCM optimizes the membership matrix by minimizing the objective function

$$J = \sum_{i=1}^{N_p}\sum_{k=1}^{C} u_{ki}^m \|p_i - v_k\|^2 \tag{20}$$

where N_P is the total number of pixels, while C is the number of clusters. $u_{ki} \in (0,1)$ denotes the membership degree of the ith pixel to the kth cluster, satisfying $\sum_{k=1}^{c} u_{ki} = 1$. The fuzziness coefficient $m > 1$ controls the clustering sharpness. Each pixel's value is denoted by p_i. $v_k$corresponds to the prototype vector (centroid) of the kth cluster, which is iteratively optimized during the FCM process.

Inspired by the enhanced FCM approach [39], which performs clustering on the gray-level histogram to reduce computational complexity, it is noted that the number of gray levels in CSAR subaperture images is typically much smaller than the total pixel count. This histogram-based clustering strategy can significantly decrease computation time. Similar to enhanced FCM, the objective function of our proposed F^{3}CM is formulated as

$$J = \sum_{l=1}^{q}\sum_{k=1}^{C} \gamma_l u_{kl}^m \|S_l - v_k\|^2 \tag{21}$$

where q is the number of gray levels in the histogram, γ_l denotes the number of pixels at gray-level l (satisfying $\sum_{l=1}^{q}\gamma_l = N_p$), and S represents the sparse matrix obtained from low-rank matrix decomposition of the CSAR subaperture image.

F^{3}CM employs the Lagrange multiplier method for iterative optimization.

The update formulas for membership degrees and cluster centers are given as follows:

$$u_{kl} = \frac{\|S_l - v_k\|^{\frac{-2}{m-1}}}{\sum_{j=1}^{C}\|S_l - v_j\|^{\frac{-2}{m-1}}} \tag{22}$$

$$v_k = \frac{\sum_{i=1}^{q}\gamma_l u_{kl}^m S_l}{\sum_{i=1}^{q}\gamma_l u_{kl}^m}. \tag{23}$$

Similar to the traditional FCM clustering algorithm, the proposed method iteratively computes the membership degree matrix $\boldsymbol{U}$ for each subaperture SAR image. In the land cover classification task, it is necessary not only to distinguish between the background and target objects but also to further separate natural terrains from man-made structures. Therefore, the proposed algorithm sets the number of cluster centers to three ($k = 3$), where the ascending order of gray levels corresponds to: background (lowest gray level), natural terrains (medium gray level), and man-made structures (highest gray level). To extract building targets, the membership matrix with the highest gray level is retained as the refined membership matrix $\boldsymbol{U}'$ representing the likelihood of pixels belonging to artificial structures.

After obtaining the membership degree matrix $\boldsymbol{U}'$ for the CSAR subaperture images, the F^{3}CM algorithm integrates multidomain features (frequency and spatial domains) to enhance classification robustness. Inspired by nonlinear

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filtering, the fusion process combines frequency- and spatial-domain features, yielding the final feature weight extraction result. The fusion formula is defined as

$$U''(x,y,n) = \frac{1}{\omega_{\text{norm}}} N\left\{M_{\text{imf}}(x,y,n)\cdot U'(x,y,n)\right\} \tag{24}$$

where ω_{norm} denotes the normalized weight coefficient and $N\{\cdot\}$ represents the nonlinear filtering.

F. Multidomain Feature Weight Fusion

The final feature map is given by

$$V_{\text{merge}}(x,y) = \sum_{n=1}^{N}\left[U''(x,y,n)L(x,y,n) - \text{Mode}(x,y)\right]^2. \tag{25}$$

The proposed method ultimately combines the aspect-domain statistical features extracted from the CSAR subaperture sequence with the multidomain fusion results to produce the final feature map.

IV. Experiments and Analysis

A. Introduction of Experimental Data

To assess the efficacy and applicability of the proposed method, experimental data from the airborne L-band CSAR system, independently developed by the National University of Defense Technology, were employed. The data acquisition took place in 2020 in Weinan, Shaanxi, China. To evaluate the method's generalization potential, diverse building types are included in the selected scenes. Using the first scene as a case study, Sections IV-B–IV-D will elaborate on the processing workflow and results analysis.

The flight configuration comprised a 1.9 km by altitude, a 2.346 km by radius, and L-band HH-polarized operation, achieving a ground resolution of 0.5 × 0.5 m. Fig. 7 illustrates the airborne SAR imaging system with platform trajectory (red curve) and region of interest (blue rectangle). Using scene one as an example, the echo data are partitioned into 84 subaperture images for analysis.

B. Qualitative Analysis

The experimental validation of the proposed algorithm is conducted using low-frequency L-band echo data for comparative analysis; three state-of-the-art CSAR image interpretation algorithms for man-made targets are selected as benchmarks, all of which have demonstrated superior performance in the adjacent C-band in recent studies.

The experimental results in Figs. 8 and 9 demonstrate the effectiveness of the proposed algorithm for building extraction in standalone-building and factory areas using L-band CSAR imagery. Compared with existing CSAR-based man-made target interpretation methods, our approach exhibits superior capability in discriminating man-made structures from natural terrain, delivering more precise building contours with fewer false alarms caused by crops or rough surfaces exhibiting moderate scattering. These findings validate the robustness of our method in complex scattering environments.

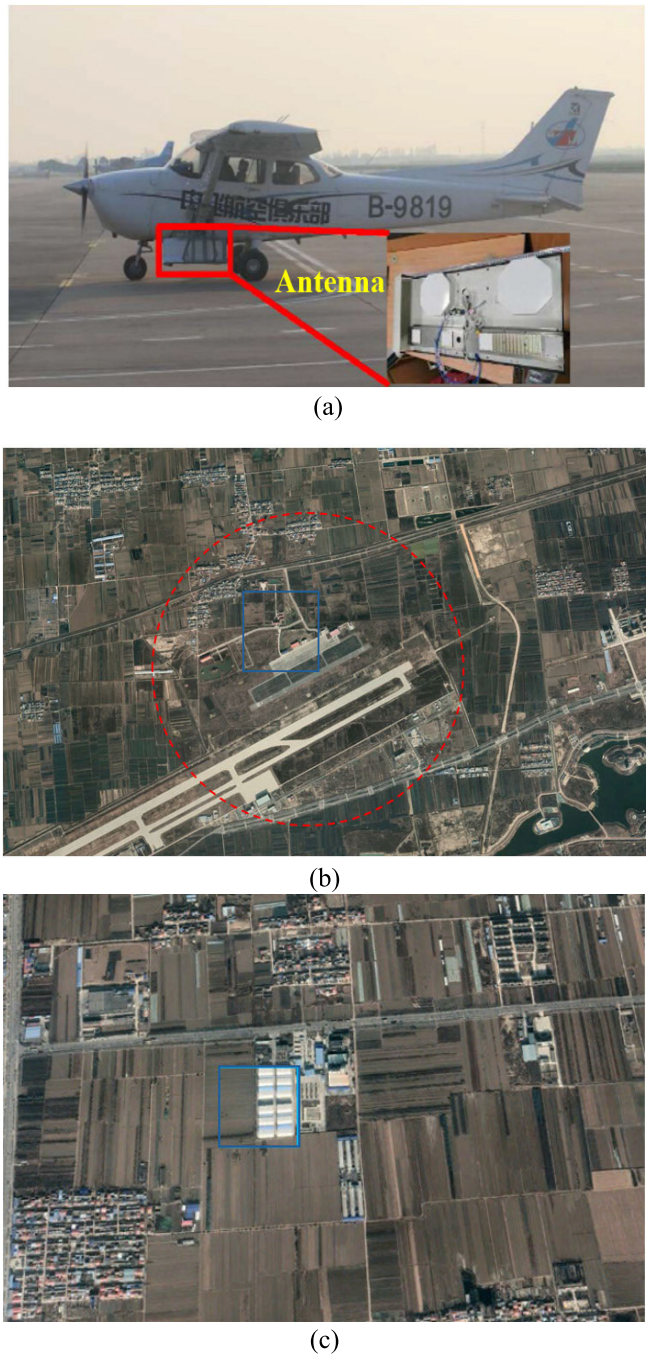


Fig. 7. CSAR experimental overview. (a) Airborne L-band CSAR system. (b) Flight trajectory and selected area of scene one. (c) Flight trajectory and selected area of scene two.

C. Quantitative Analysis

To more intuitively demonstrate the advantages of the algorithm, a comprehensive evaluation was conducted using a variety of performance metrics: detection rate, false discovery rate, accuracy, and the Dice coefficient.

The classification performance is quantitatively evaluated using four fundamental metrics in the confusion matrix: 1) true positives (TPs) represent the number of correctly identified building pixels, where both the ground truth and predicted labels are buildings; 2) false negatives (FNs) denote the

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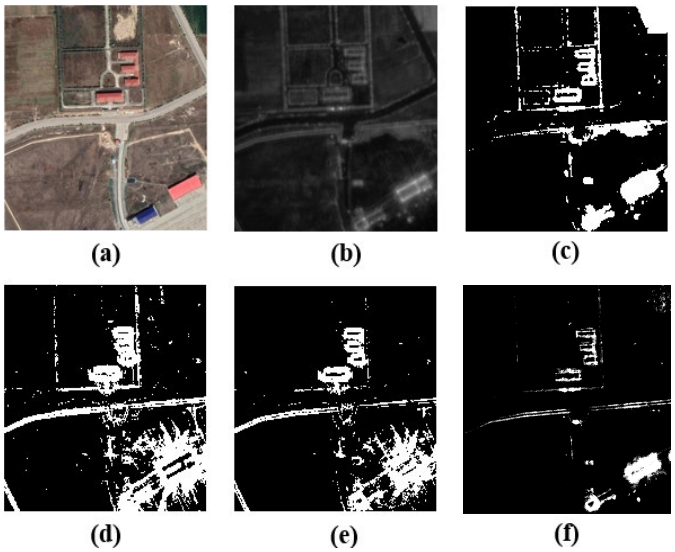


Fig. 8. Extraction results of scene one using the L-band data. (a) Optical image. (b) CSAR image. (c) Method in [33]. (d) Method in [35]. (e) Method in [36]. (f) Proposed method.

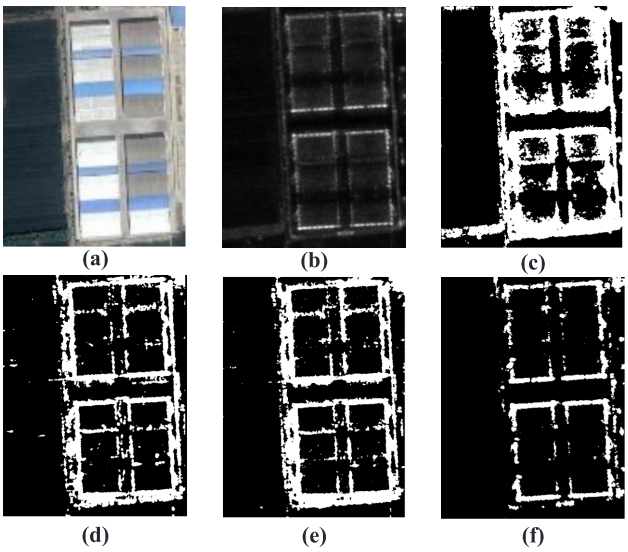


Fig. 9. Extraction results of scene two using the L-band data. (a) Optical image. (b) CSAR image. (c) Method in [33]. (d) Method in [35]. (e) Method in [36]. (f) Proposed method.

number of undetected building pixels (type II error), where the ground truth is building but the prediction is nonbuilding; 3) false positives (FPs) indicate the number of nonbuilding pixels erroneously classified as buildings (type I error), where the ground truth is nonbuilding but the prediction is building; and 4) true negatives (TNs) correspond to the number of correctly classified nonbuilding pixels, where both labels agree on nonbuilding.

Their definitions are as follows:

$$\mathrm{DR} = \frac{\mathrm{TP}}{\mathrm{FN} + \mathrm{TP}} \tag{26}$$

$$\mathrm{FDR} = \frac{\mathrm{FP}}{\mathrm{TP} + \mathrm{FP}} \tag{27}$$

$$\mathrm{AC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{FN} + \mathrm{FP} + \mathrm{TP} + \mathrm{TN}} \tag{28}$$

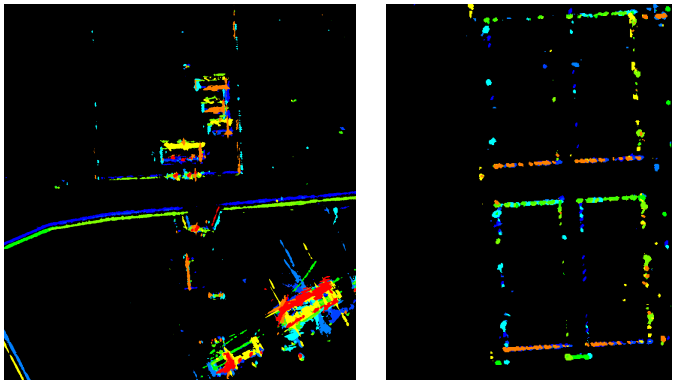


Fig. 10. Extraction results of strong scattering points across different subaperture images. (Different colors represent the aircraft rotating through different angle ranges during flight.)

$$\mathrm{Dice} = \frac{2 \times \mathrm{TP}}{\mathrm{FN} + \mathrm{FP} + 2 \times \mathrm{TP}}. \tag{29}$$

The experimental results demonstrate that the proposed method achieves a balanced and robust performance compared to existing approaches. While the method in [33] attains the highest detection rate (DR: 96.49%), the proposed method significantly reduces the false discovery rate (FDR: 33.57%) by nearly half compared to the next best method (method in [35]: 55.83%), indicating superior reliability in minimizing spurious detections. Moreover, the proposed method achieves the highest accuracy (AC: 98.20%) and Dice coefficient (Dice: 78.00%), outperforming all baselines by a considerable margin, which underscores its enhanced segmentation precision and overall correctness. Although the DR of the proposed method (94.44%) is slightly lower than that of [33], the substantial improvements in FDR, AC, and Dice suggest a favorable tradeoff, making the proposed method more suitable for applications where precision and robustness are critical (Table II).

The proposed algorithm achieves groundbreaking advancements in processing CSAR subaperture image sequences, with its core innovation residing in the development of a dual-module collaborative architecture integrating SFMD-SSPE and LASPE. Through the SFMD-SSPE module, the algorithm accurately captures the persistent strong scattering signatures characteristic of artificial structures, while the LASPE module effectively deciphers target anisotropic scattering configurations, demonstrating superior performance in discriminating scattering properties between natural terrains and man-made targets. Comparative experiments validate that our method significantly outperforms state-of-the-art CSAR feature extraction techniques in both discrimination accuracy and robustness, particularly excelling in artificial target detection within low-frequency bands. This work establishes a novel technical framework for low-frequency band CSAR image interpretation, offering enhanced capabilities for precise target characterization.

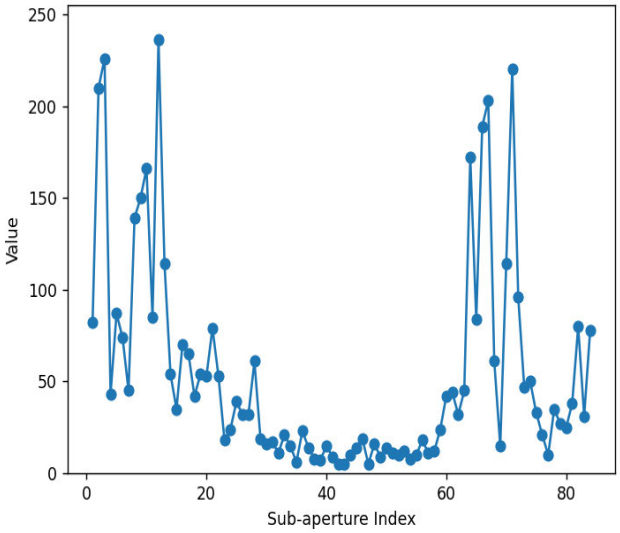
D. Extended Analysis and Discussion

As illustrated in Fig. 10, strong scattering points extracted from subaperture images within different aspect angle ranges

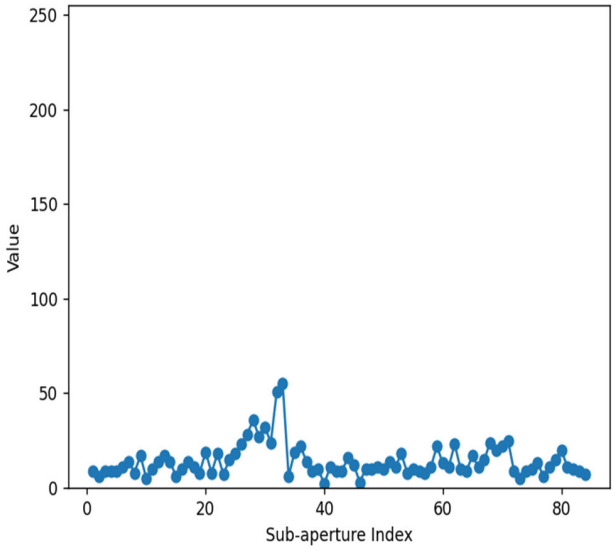
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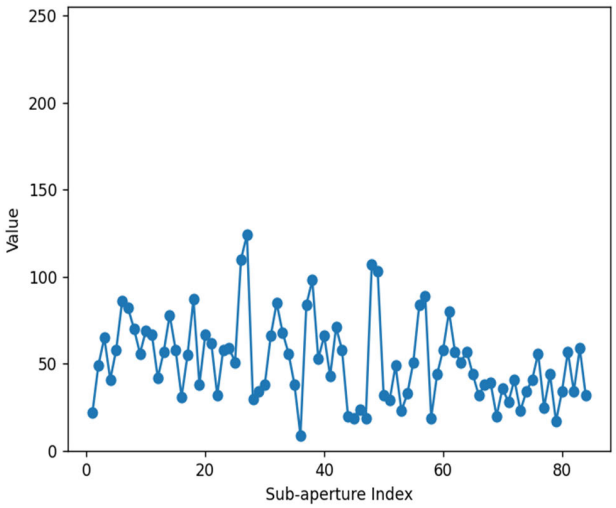
(a)



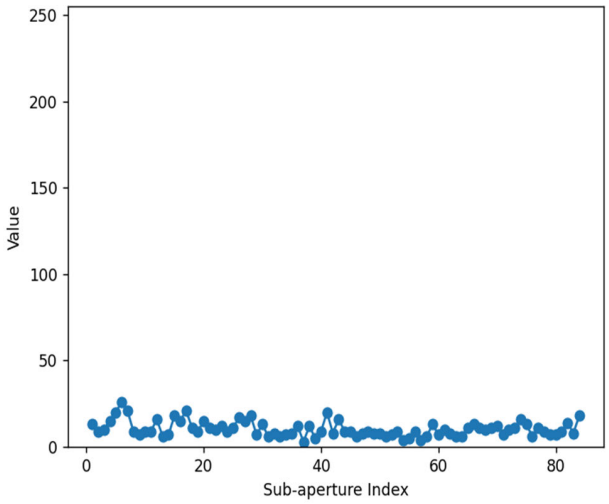
(b)



(c)



(d)



(e)

Fig. 11. Scattering characteristics analysis of four typical terrain features in subaperture image sequences. (a) Location selection of measurement points. (b) Scattering amplitude of building structure in subaperture sequence. (c) Scattering amplitude of rough terrain surface in subaperture sequence. (d) Scattering amplitude of agricultural crops in subaperture sequence. (e) Scattering amplitude of paved roadway in subaperture sequence.

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can be distinguished using distinct colors. The results demonstrate noticeable variations in strong scattering points across subaperture images in the CSAR image sequence, which aligns with our previous conclusion that man-made structures exhibit strong scattering characteristics only in specific subaperture ranges.

For an in-depth analysis of scattering characteristics from typical terrain features in CSAR subaperture imagery, Fig. 11(a) demonstrates the experimental configuration with four distinct terrain features analyzed using measured airborne L-band radar returns: a building structure at approximate location A, an uneven terrain surface at approximate location B, agricultural crop fields at approximate location C, and a paved roadway at approximate location D.

As illustrated in Fig. 11(b)–(e), the building structure exhibits elevated scattering amplitudes in partial consecutive subapertures while demonstrating reduced magnitudes in remaining sequences. In contrast, the rough terrain surface maintains near-zero backscattering across most subapertures, with only marginal scattering observed in isolated apertures. Although agricultural crops do not manifest strong scattering behavior, they generate measurable backscatter at L-band frequencies, displaying stochastic scattering patterns throughout the subaperture sequence. The paved roadway yields extremely weak backscattering signals across nearly all subapertures, leading to consistently minimal signal amplitudes in each subaperture.

To investigate the influence of subaperture sequence selection on the final extraction results, downsampling was performed on the subaperture image sequence. In this study, downsampling specifically refers to uniformly selecting subaperture images at fixed intervals from the original 84 subaperture sequences, taking scene one as an example. For instance, selecting every second subaperture image reduces the sequence to 42 aspect angles, and so forth. This approach allows systematic evaluation of how subaperture sampling density affects scattering point extraction performance.

As illustrated in Figs. 12 and 13, ablation studies were conducted with downsampled subaperture sequences to further validate the effectiveness of the two proposed feature extraction modules—SFMD-SSPE and LASPE—which are specifically designed for capturing distinct scattering characteristics of buildings in subaperture images. When applied individually for building extraction, both LASPE and SFMD-SSPE modules demonstrated consistent robustness in extraction performance across varying numbers of subapertures.

As illustrated in Fig. 14, the proposed algorithm has demonstrated promising performance in building extraction tasks at the L-band. To investigate its sensitivity to radar frequency, the algorithm was further validated using the Ku-band echo data. Experimental results indicate that the algorithm maintains robust performance in the Ku-band, achieving accurate building extraction with notably few false alarms.

As illustrated in Fig. 15, this demonstrates the extraction results of the proposed algorithm under different downsampling intervals for the subaperture sequence. When the sampling interval is relatively small, the proposed algorithm

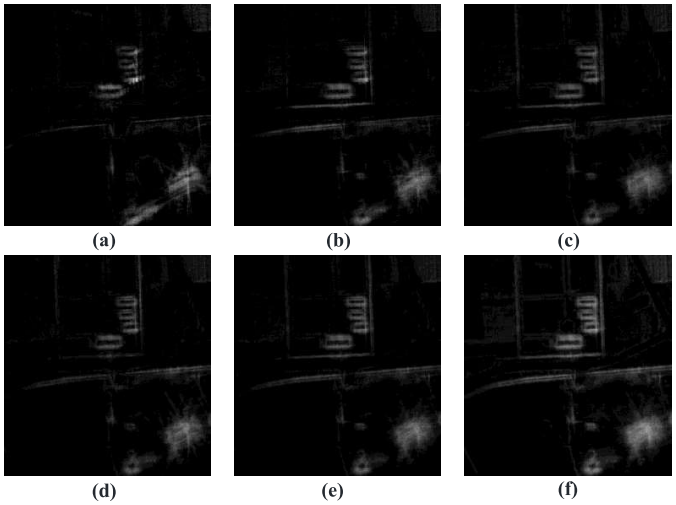


Fig. 12. LASPE results under subaperture sequence downsampling. (a) Extraction result for seven aspects. (b) Extraction result for 14 aspects. (c) Extraction result for 21 aspects. (d) Extraction result for 28 aspects. (e) Extraction result for 42 aspects. (f) Extraction result for 84 aspects.

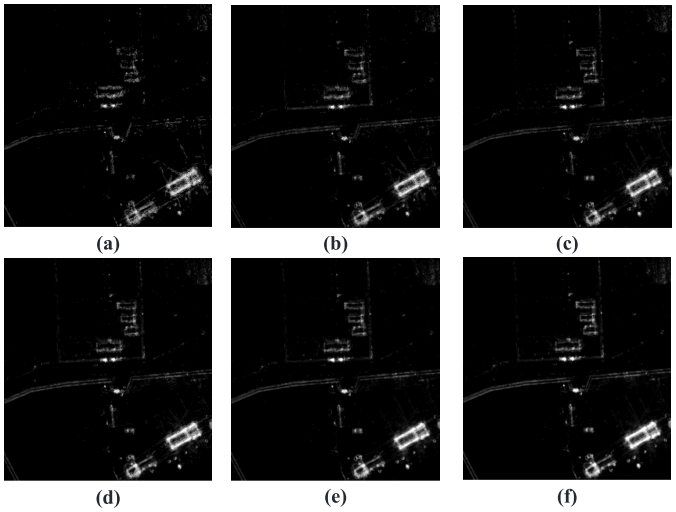


Fig. 13. SFMD-SSPE results under subaperture sequence downsampling. (a) Extraction result for seven aspects. (b) Extraction result for 14 aspects. (c) Extraction result for 21 aspects. (d) Extraction result for 28 aspects. (e) Extraction result for 42 aspects. (f) Extraction result for 84 aspects.

maintains satisfactory extraction performance, as evidenced by the extraction results of 42 aspects. However, with larger sampling intervals, the absence of certain subaperture images that characterize strong building scatterers may lead to missed detections of pixels belonging to the building category.

Table III demonstrates the superior performance of the proposed method across different subaperture configurations when compared to existing approaches. Using only seven subapertures, the method achieves competitive results while significantly reducing computational complexity without excessively compromising accuracy. With 84 subapertures, the method outperforms all benchmarks, reaching a peak Dice coefficient of 78.00%, an accuracy of 98.20%, and the lowest false discovery rate of 33.57%.

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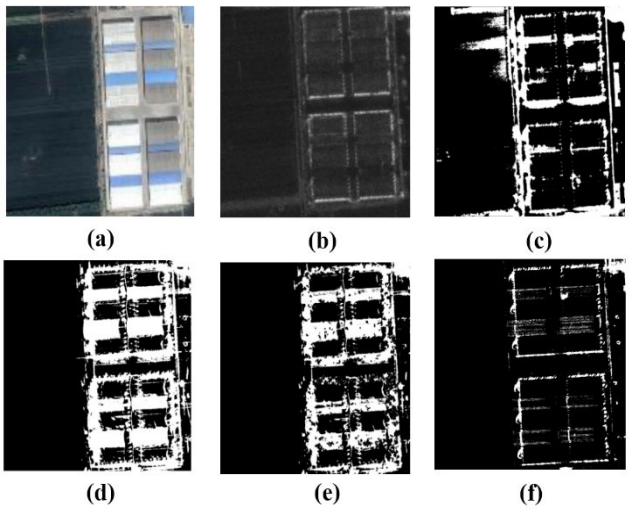


Fig. 14. Extraction results of scene two using the Ku-band data. (a) Optical image. (b) CSAR image. (c) Method in [33]. (d) Method in [35]. (e) Method in [36]. (f) Proposed method.

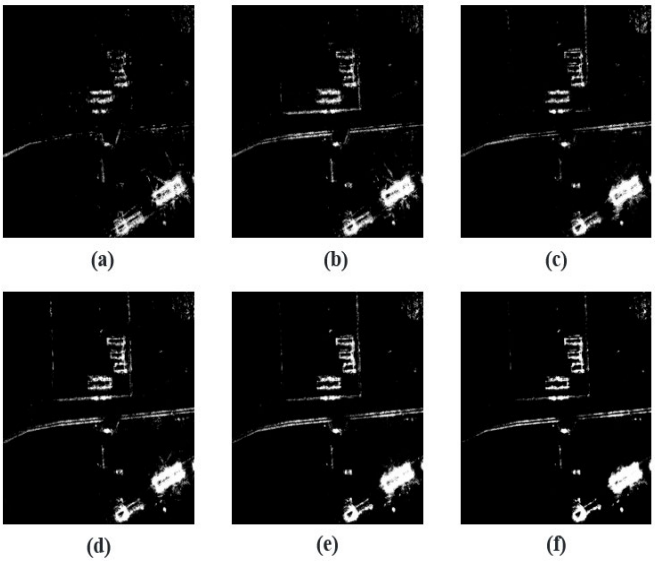


Fig. 15. Building extraction results under subaperture sequence downsampling. (a) Extraction result for seven aspects. (b) Extraction result for 14 aspects. (c) Extraction result for 21 aspects. (d) Extraction result for 28 aspects. (e) Extraction result for 42 aspects. (f) Extraction result for 84 aspects.

TABLE II
EVALUATION METRICS FOR SCENE ONE

Method	Evaluation Metrics			
	DR(%)	*FDR(%)*	*AC(%)*	*Dice(%)*
The method in [33]	**96.49**	58.91	95.21	57.64
The method in [35]	94.57	55.83	95.78	60.22
The method in [36]	91.27	62.72	94.52	52.94
The proposed method	94.44	**33.57**	**98.20**	**78.00**

As the number of downsampled subaperture images increases, the proposed method demonstrates consistent improvements in detection rate coupled with reductions in the FDR across a wide range of aspect counts (from 7 to 84), highlighting its robustness to sparse sampling conditions. Simultaneously, the Dice coefficient demonstrates a consistent upward trend. Notably, using only seven subaperture images, our algorithm still achieves significantly better extraction results than the comparative methods.

TABLE III
EXTRACTION RESULTS AFTER ALGORITHM DOWNSAMPLING

Method	Number of Sub-apertures	Evaluation Metrics			
		DR(%)	*FDR(%)*	*AC(%)*	*Dice(%)*
The method in [33]	84	96.49	58.91	95.21	57.64
The method in [35]	84	94.57	55.83	95.78	60.22
The method in [36]	84	91.27	62.72	94.52	52.94
The proposed method	7	85.25	52.07	96.38	61.36
	14	95.51	42.40	97.48	71.86
	21	90.63	39.83	97.66	72.32
	28	95.77	44.28	97.29	70.45
	42	**96.54**	40.96	97.62	73.27
	84	94.44	**33.57**	**98.20**	**78.00**

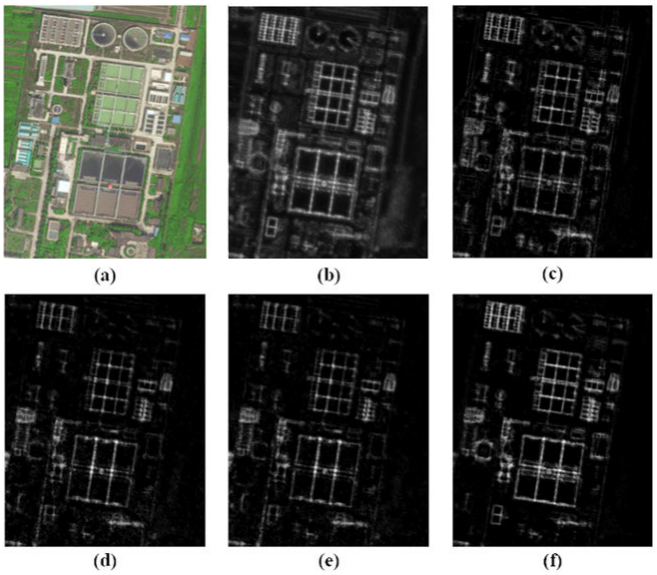


Fig. 16. Extraction results under different polarization modes. (a) Optical image. (b) CSAR image. (c) HH-polarization extraction result. (d) HV-polarization extraction result. (e) VH-polarization extraction result. (f) VV-polarization extraction result.

As shown in Fig. 16, to further validate the robustness of the proposed algorithm in extracting buildings under different polarization modes and complex backgrounds, a scene containing various types of artificial features in the L-band is selected. Based on the subaperture sequences of different polarization modes, it can be observed that the extraction results under HH- and VV-polarizations are similar, while those under HV- and VH-polarizations are also comparable. Furthermore, the building extraction performance under HH- and VV-polarizations is superior to that under HV- and VH-polarizations, yielding more complete building contours.

As shown in Fig. 17, the experimental results using one-third subaperture images reveal that the proposed algorithm can only extract building pixels exhibiting strong scattering

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Fig. 17. Building extraction results using limited one-third subaperture images. (a) Extraction results using the first one-third subaperture sequence. (b) Extraction results using the middle one-third subaperture sequence. (c) Extraction results using the last one-third subaperture sequence.

responses within the observed subaperture image sequence. Missed detections may occur for pixels without strong scattering in the selected observation angles, which stems from the imaging mechanism of CSAR: building pixels typically manifest strong scattering only within specific limited angle ranges.

This limitation highlights a key drawback of our method—instability in extraction results under constrained observation angles. Future research could address this issue through scattering property inference or contour completion techniques to enhance robustness.

V. Conclusion

In this article, two novel methods are proposed, SFMD-SSPE and LASPE, to address the challenges of building extraction in CSAR subaperture image sequences. The SFMD-SSPE method effectively extracts strong scattering points through spatial–frequency matrix decomposition, while LASPE captures anisotropic scattering characteristics by analyzing the statistical properties of low-rank CSAR subaperture sequences. Experimental results on airborne L-band CSAR data demonstrate that our approach achieves higher accuracy in building extraction compared to existing CSAR-based methods for man-made target detection.

However, the performance of the proposed methods is affected by limited aspect angle observations, leading to incomplete building contours in such scenarios. Future research will focus on two key directions: 1) developing techniques to infer missing angular data from limited observations and 2) improving building contour completion algorithms specifically tailored for CSAR imagery with restricted aspect angles. These advancements could further enhance the robustness of CSAR-based building extraction in practical applications.

References

[1] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 1, pp. 6–43, Mar. 2013.

[2] B. Zou, W. Li, and L. Zhang, "Built-up area extraction using high-resolution SAR images based on spectral reconfiguration," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 8, pp. 1391–1395, Aug. 2021.

[3] K. Cho, S.-E. Park, J.-H. Cho, H. Moon, and S.-H. Han, "Automatic urban area extraction from SAR image based on morphological operator," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 831–835, May 2021.

[4] W. Wu, H. Guo, and X. Li, "Urban area SAR image man-made target extraction based on the product model and the time–frequency analysis," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 3, pp. 943–952, Mar. 2015.

[5] M. Shahzad, M. Maurer, F. Fraundorfer, Y. Wang, and X. X. Zhu, "Buildings detection in VHR SAR images using fully convolution neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1100–1116, Feb. 2019.

[6] F. Argenti, A. Lapini, T. Bianchi, and L. Alparone, "A tutorial on speckle reduction in synthetic aperture radar images," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 3, pp. 6–35, Sep. 2013.

[7] G. Fornaro, F. Lombardini, A. Pauciullo, D. Reale, and F. Viviani, "Tomographic processing of interferometric SAR data: Developments, applications, and future research perspectives," *IEEE Signal Process. Mag.*, vol. 31, no. 4, pp. 41–50, Jul. 2014.

[8] O. Pappas, A. Achim, and D. Bull, "Superpixel-level CFAR detectors for ship detection in SAR imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 9, pp. 1397–1401, Sep. 2018.

[9] G. Gao, S. Gao, J. He, and G. Li, "Adaptive ship detection in hybrid-polarimetric SAR images based on the power–entropy decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5394–5407, Sep. 2018.

[10] J. Kang, Z. Wang, and R. Zhu, "Supervised contrastive learning regularized high-resolution synthetic aperture radar building footprint generation," *J. Radars*, vol. 11, no. 1, pp. 157–167, 2022.

[11] E. Simonetto, H. Oriot, and R. Garello, "Rectangular building extraction from stereoscopic airborne radar images," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 10, pp. 2386–2395, Oct. 2005.

[12] L. Zhao, X. Zhou, and G. Kuang, "Building detection from urban SAR image using building characteristics and contextual information," *EURASIP J. Adv. Signal Process.*, vol. 2013, no. 1, Dec. 2013, Art. no. 125.

[13] X. Guanlei, W. Xiaotong, and X. Xiaogang, "On analysis of bi-dimensional component decomposition via BEMD," *Pattern Recognit.*, vol. 45, no. 4, pp. 1617–1626, Apr. 2012.

[14] L. Chen, D. An, and X. Huang, "A backprojection-based imaging for circular synthetic aperture radar," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 8, pp. 3547–3555, Aug. 2017.

[15] L. Chen, D. An, and X. Huang, "Resolution analysis of circular synthetic aperture radar noncoherent imaging," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 1, pp. 231–240, Jan. 2020.

[16] L. Chen, D. An, X. Huang, and Z. Zhou, "A 3D reconstruction strategy of vehicle outline based on single-pass single-polarization CSAR data," *IEEE Trans. Image Process.*, vol. 26, no. 11, pp. 5545–5554, Nov. 2017.

[17] D. An, L. Chen, F. Dong, X. Huang, and Z. Zhi-Min, "Study of the airborne circular synthetic aperture radar imaging technology," *J. Radars*, vol. 9, no. 2, pp. 221–242, Feb. 2020.

[18] D. Feng, D. An, L. Chen, and X. Huang, "Holographic SAR tomography 3-D reconstruction based on iterative adaptive approach and generalized likelihood ratio test," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 305–315, Jan. 2021.

[19] X. Li, G. Zhang, C. Yin, Y. Wu, and X. Shen, "A novel shadow and layover segmentation network for multi-angle SAR images fusion," *IEEE Access*, vol. 10, pp. 117770–117781, 2022.

[20] S. Palm, R. Sommer, D. Janssen, A. Tessmann, and U. Stilla, "Airborne circular W-band SAR for multiple aspect urban site monitoring," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6996–7016, Sep. 2019.

[21] Y. Luo, D. An, W. Wang, L. Chen, and X. Huang, "Local road area extraction in CSAR imagery exploiting improved curvilinear structure detector," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022, doi: 10.1109/TGRS.2022.3172227.

[22] F. Teng, Y. Lin, Y. Wang, W. Shen, S. Feng, and W. Hong, "Multi-angular SAR statistical properties analysis and man-made target detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2020, pp. 782–785.

[23] Y. Luo, D. An, W. Wang, and X. Huang, "Improved ROEWA SAR image edge detector based on curvilinear structures extraction," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 4, pp. 631–635, Apr. 2020.

[24] Y. Luo, D. An, and X. Huang, "An improved PGA method combined with CSAR time-domain imaging processing," *J. Signal Process.*, vol. 33, no. 9, pp. 1153–1161, 2017.

[25] F. Teng, Y. Lin, S. Feng, and W. Hong, "A man-made target detection method based on multi-angular phase characteristic," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 5024–5027.

[26] Y. Lin, W. Hong, W. Tan, Y. Wang, and M. Xiang, "Airborne circular SAR imaging: Results at P-band," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2012, pp. 5594–5597.

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[27] Y. Li, Q. Yin, Y. Lin, and W. Hong, "Anisotropy scattering detection from multiaspect signatures of circular polarimetric SAR," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 10, pp. 1575–1579, Oct. 2018.
[28] F. Xue et al., "Analysis of azimuthal variations using multi-aperture polarimetric entropy with circular SAR images," *Remote Sens.*, vol. 10, no. 1, p. 123, Jan. 2018.
[29] X. Tan, D. An, L. Chen, Y. Luo, Z. Zhou, and D. Zhao, "An effective method of bridge detection based on polarimetric CSAR," in *Proc. 21st Int. Radar Symp. (IRS)*, Oct. 2020, pp. 131–134.
[30] F. Teng, W. Hong, and Y. Lin, "Aspect entropy extraction using circular SAR data and scattering anisotropy analysis," *Sensors*, vol. 19, no. 2, p. 346, Jan. 2019.
[31] Q. Liu, W. Yu, and W. Hong, "Vehicle detection in multi-aspect SAR images based on improved GOFRO," *J. Radars*, vol. 12, no. 5, pp. 1081–1096, 2023.
[32] Y. Zhao, Y. Lin, Y. P. Wang, W. Hong, and L. Yu, "Target multi-aspect scattering sensitivity feature extraction based on circular-SAR," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 2722–2725.
[33] X. Yue, F. Teng, Y. Lin, and W. Hong, "A man-made target extraction method based on scattering characteristics using multiaspect SAR data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11699–11712, 2021.
[34] J. Zhao, D. An, and L. Chen, "A novel method for building contour extraction based on CSAR images," *Remote Sens.*, vol. 15, no. 14, p. 3463, Jul. 2023.
[35] F. Teng et al., "An anisotropic scattering analysis method based on likelihood ratio using circular SAR data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 477–480.
[36] F. Teng, Y. Lin, Y. Wang, W. Shen, S. Feng, and W. Hong, "An anisotropic scattering analysis method based on the statistical properties of multi-angular SAR images," *Remote Sens.*, vol. 12, no. 13, p. 2152, Jul. 2020.
[37] Y. Li, L. Chen, D. An, X. Huang, and D. Feng, "A novel method for extracting geometric parameter information of buildings based on CSAR images," *Int. J. Remote Sens.*, vol. 43, no. 11, pp. 4117–4133, Jun. 2022.
[38] D. Yang, X. Yang, G. Liao, and S. Zhu, "Strong clutter suppression via RPCA in multichannel SAR/GMTI system," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2237–2241, Nov. 2015.
[39] L. Szilagyi, Z. Benyo, S. M. Szilagyi, and H. S. Adam, "MR brain image segmentation using an enhanced fuzzy C-means algorithm," in *Proc. 25th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, Sep. 2003, pp. 724–726.



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